

SESSION 2 • LECTURE 2

Thermodynamic Equilibrium & the Quasi-static Process

Detailed lecture notes — equilibrium conditions, quasi-static idealisation, reversibility, solved numericals

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Learning objectives. After this session you should be able to: (i) state the three conditions of thermodynamic equilibrium; (ii) explain why properties are defined only at equilibrium; (iii) describe a quasi-static process and justify its use; (iv) distinguish reversible and irreversible processes; (v) apply a force balance to find the equilibrium pressure of a confined gas.

1. Thermodynamic Equilibrium

A system is in **thermodynamic equilibrium** when its properties do not change with time *and* no changes occur when the system is isolated from its surroundings. This requires three separate equilibria simultaneously:

Equilibrium	Condition inside the system	Driving potential removed	Violation example
Thermal	Uniform temperature throughout; no heat flow	$\Delta T = 0$	Hot coffee cooling in a room
Mechanical	No unbalanced forces; uniform pressure	$\Delta p = 0$	Gas expanding against a suddenly released piston
Chemical	No reaction or diffusion; uniform composition	$\Delta \mu = 0$	Burning fuel; salt dissolving in water

If any driving potential (temperature difference, pressure difference, chemical potential difference) exists, a spontaneous change proceeds until the potential vanishes — only then is the system in equilibrium.

KEY POINT Properties such as p and T have single, well-defined values only at equilibrium. A state can be plotted as a point on a property diagram only if it is an equilibrium state.

2. The Quasi-static Process

Real processes pass through non-equilibrium conditions: during a rapid expansion the pressure near the piston is lower than in the bulk, so no single p describes the gas. Classical thermodynamics escapes this difficulty with an idealisation.

A **quasi-static** (quasi-equilibrium) process is one carried out so slowly that the system passes through a continuous succession of equilibrium states; the departure from equilibrium is infinitesimal at every instant.

The sand-grain thought experiment

1. A gas is confined by a piston loaded with a pile of sand; the gas is in equilibrium at pressure p_1 .
2. Remove one grain: the pressure imbalance is infinitesimal; the piston rises a tiny amount and equilibrium is re-established almost instantly.
3. Repeat grain by grain. At every stage the gas has definite p , V , T ; the full path 1→2 is a continuous curve of equilibrium states.
4. If instead the entire load were removed at once, the gas would surge irreversibly through undefined intermediate conditions — only the end states could be marked.

Aspect	Quasi-static	Non-quasi-static (rapid)
Intermediate states	All equilibrium; fully defined	Undefined; system non-uniform
p - V representation	Continuous solid curve	End points only (dashed line)
Work evaluation	$W = \int p \, dV$ valid	$\int p \, dV$ meaningless
Reversibility	Necessary condition for reversibility	Always irreversible

3. Reversible and Irreversible Processes

A **reversible process** can be reversed so that both system and surroundings return exactly to their initial states, leaving no trace

anywhere. It must be (i) quasi-static and (ii) free of all dissipative effects — friction, unrestrained expansion, mixing, and heat transfer across a finite temperature difference. Reversible processes yield the **maximum work output** (or require minimum work input) and serve as the benchmark against which real machines are judged throughout Units II-V.

- Common irreversibilities: solid and fluid friction, free expansion, throttling, mixing of gases, heat transfer through a finite ΔT , electrical resistance, inelastic deformation.
- All real processes are irreversible; well-lubricated, slow processes approach reversibility.

4. Worked Examples

WORKED EXAMPLE 2.1 — Equilibrium pressure of a loaded piston

A vertical piston-cylinder contains gas. The frictionless piston has mass 40 kg and face area 0.01 m^2 ; atmospheric pressure is 100 kPa; $g = 9.81 \text{ m/s}^2$. Find the gas pressure required to hold the piston in equilibrium.

Solution.

1. Free body of the piston: upward force $p_{\text{gas}}A$; downward forces $p_{\text{atm}}A + mg$.
2. Equilibrium: $p_{\text{gas}}A = p_{\text{atm}}A + mg \Rightarrow p_{\text{gas}} = p_{\text{atm}} + mg/A$.
3. $mg/A = (40 \times 9.81)/0.01 = 39,240 \text{ Pa} = 39.24 \text{ kPa}$.
4. $p_{\text{gas}} = 100 + 39.24 = 139.24 \text{ kPa}$.

ANSWER $p_{\text{gas}} \approx 139.2 \text{ kPa}$

Note: The loaded piston fixes the pressure. Heating this gas produces a constant-pressure (isobaric) process — the basis of $W = p\Delta V$ in Lecture 3.

WORKED EXAMPLE 2.2 — Piston with a spring (extension)

In the cylinder of Example 2.1, a spring now also presses on the piston with a force of 600 N. Find the new equilibrium gas pressure.

Solution.

1. Additional downward force per unit area = $F_{\text{spring}}/A = 600/0.01 = 60,000 \text{ Pa} = 60 \text{ kPa}$.
2. $p_{\text{gas}} = p_{\text{atm}} + mg/A + F_{\text{spring}}/A = 100 + 39.24 + 60 = 199.24 \text{ kPa}$.

ANSWER $p_{\text{gas}} \approx 199.2 \text{ kPa}$

Note: As the gas expands and compresses the spring further, the spring force — and hence the gas pressure — rises linearly with volume: p - V path is a straight line.

WORKED EXAMPLE 2.3 — Identifying equilibrium

State, with reasons, whether each is an equilibrium state: (a) a cup of tea at 70°C sitting in a 30°C room; (b) a gas mixture mid-way through combustion; (c) air in a sealed, long-undisturbed room at uniform 25°C and 1 atm .

Solution.

1. (a) No — a finite $\Delta T = 40^\circ\text{C}$ with the surroundings drives heat flow: thermal equilibrium absent.
2. (b) No — chemical reaction is in progress: chemical equilibrium absent.
3. (c) Yes — uniform T and p , no reactions, no unbalanced forces: all three equilibria satisfied.

ANSWER (a) No (b) No (c) Yes

5. Practice Problems (answers given)

#	Problem	Answer
P1	A frictionless piston of mass 60 kg and area 0.015 m^2 rests on a gas; $p_{\text{atm}} = 101 \text{ kPa}$. Find the gas pressure.	$\approx 140.2 \text{ kPa}$
P2	Why can a rapid compression not be shown as a continuous curve on a p - V diagram? Answer in two lines.	Intermediate states are non-equilibrium; p is undefined
P3	Give two dissipative effects that make a process irreversible even if it is slow.	e.g. friction; heat transfer across finite ΔT

6. Summary

- Thermodynamic equilibrium = thermal + mechanical + chemical equilibrium, simultaneously.
- A quasi-static process is a succession of equilibrium states, achieved by infinitely slow change.

- Only quasi-static paths can be drawn as continuous curves and integrated ($\int p \, dV$).
- Reversible = quasi-static + no dissipation; it is the ideal limit of real processes.
- Force balance on a loaded piston: $p_{\text{gas}} = p_{\text{atm}} + mg/A$ (+ spring force/A).

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 1 (Introduction). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.